



Rajshahi University of Engineering & Technology

Department of Mechatronics Engineering

LAB REPORTS

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Course Title: **Hydraulic and Pneumatic Control Sessional**

Experiment No: **04**

Name of the Experiment: Study of a Single Acting Cylinder Using Logic IC Gate

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Experiment Name: Study of a Single Acting Cylinder Using Logic IC Gate

Objectives

- To understand the working principle of a single acting pneumatic cylinder.
- To study and apply various logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR).
- To design and implement electro-pneumatic circuits controlled by logic gates.
- To observe how logical inputs control the movement of a single acting cylinder.
- To demonstrate logical decision-making in practical automation systems.

Theory

A single acting pneumatic cylinder is a linear actuator that uses compressed air to move a piston in one direction—typically to extend. A spring mechanism inside the cylinder returns the piston to its original position once the air is released. These cylinders usually operate at an air pressure of 80–100 PSI and are commonly controlled with a 3/2-way valve [1].

When compressed air enters the inlet, it pushes the piston forward. When the air is released and vented, the spring retracts the piston. These cylinders are cost-effective, air-efficient, and ideal for applications such as clamping, ejecting, or lifting lightweight objects. However, they generate force in only one direction and are limited by the return spring strength [2].

In pneumatic systems, logical functions can be replicated using air flow and control valves. High air pressure (flow) represents a logic 1. No air flow (low pressure) represents a logic 0.

AND Gate (Series Connection)

- Pneumatic Representation: Two 3/2 valves connected in series. Air passes only if both valves are open.
- Logical Behavior: Output is 1 only if both inputs are 1.

Input A	Input B	Output
0	0	0
0	1	0
1	0	0
1	1	1

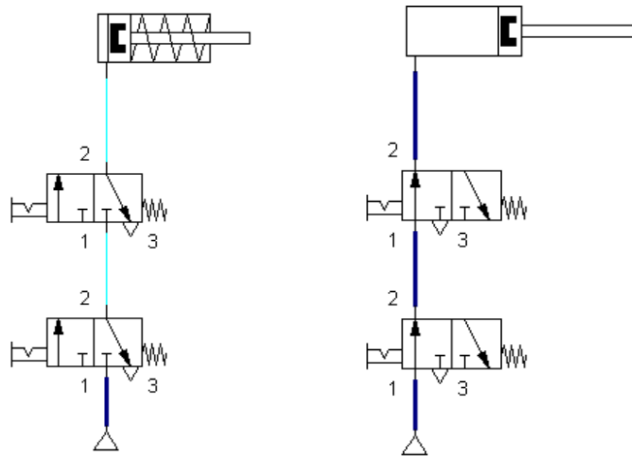


Figure 4.1: AND Gate Series Connection in (0,0) & (1,1) states

OR Gate (Parallel Connection)

- Pneumatic Representation: Two valves connected in parallel using a shuttle valve.
- Logical Behavior: Output is 1 if at least one input is 1.

Input A	Input B	Output
0	0	0
0	1	1
1	0	1
1	1	1

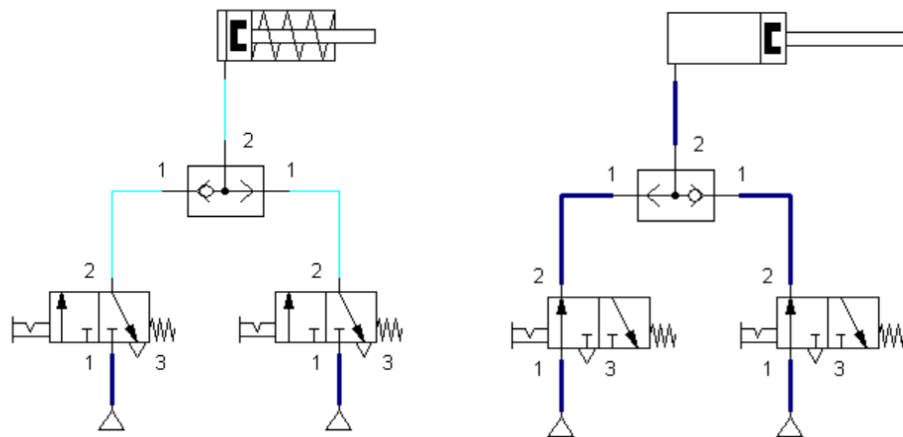


Figure 4.2: OR Gate Parallel Connection in (0,0) & (1,1) states

NOT Gate (Inverter)

- Pneumatic Representation: Inverts the logic signal—air is blocked when the switch is pressed.
- Logical Behavior: Output is 1 when input is 0, and 0 when input is 1.

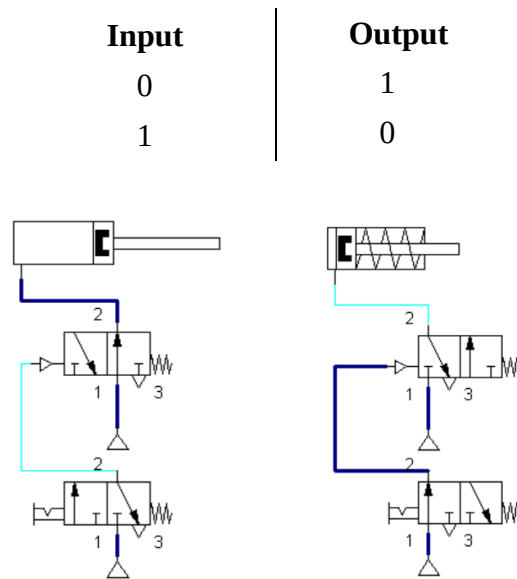


Figure 4.3: NOT Gate in Switch OFF and ON states

NAND Gate

- Pneumatic Representation: Combination of shuttle valve and normally open 3/2 valves.
- Logical Behavior: Output is 1 unless both inputs are 1.

Input A	Input B	Output
0	0	1
0	1	1
1	0	1
1	1	0

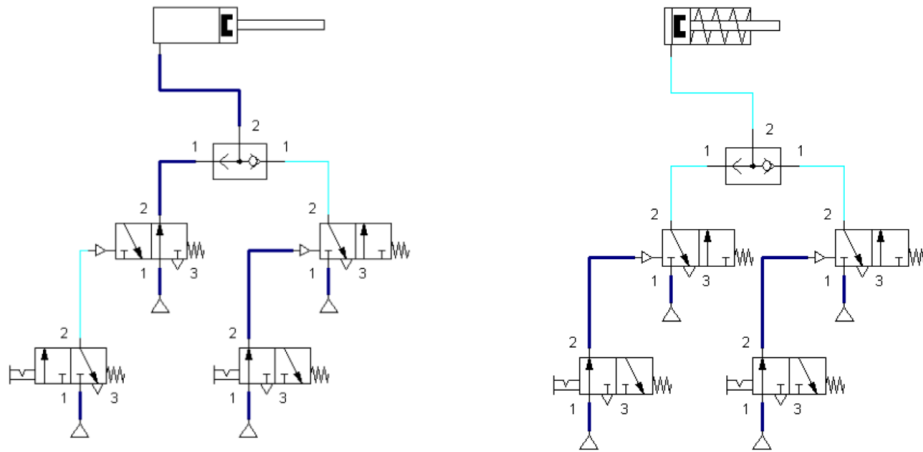


Figure 4.4: NAND Gate in (0,1) & (1,1) states

NOR Gate

- **Pneumatic Representation:** Created using an AND configuration with pressure valves and normally closed 3/2 DCVs.
- **Logical Behavior:** Output is 1 only when both inputs are 0.

Input A	Input B	Output
0	0	1
0	1	0
1	0	0
1	1	0

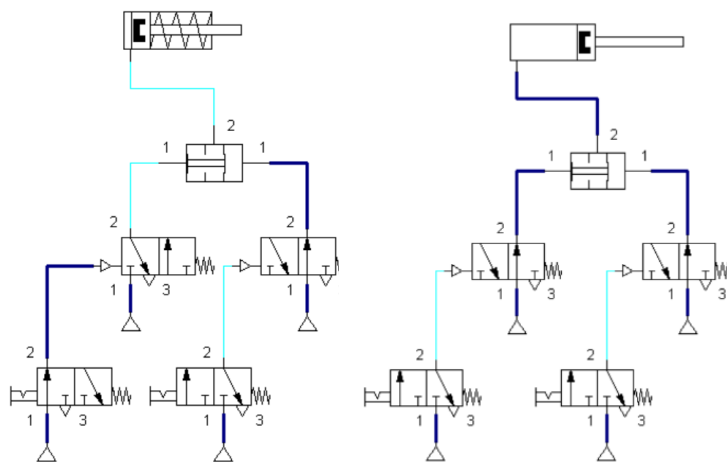


Figure 4.5: NOR Gate in (1,0) & (0,0) states

XOR Gate (Exclusive OR)

- Pneumatic Representation: Air flows only when one valve is open.
- Logical Behavior: Output is 1 if the inputs differ.

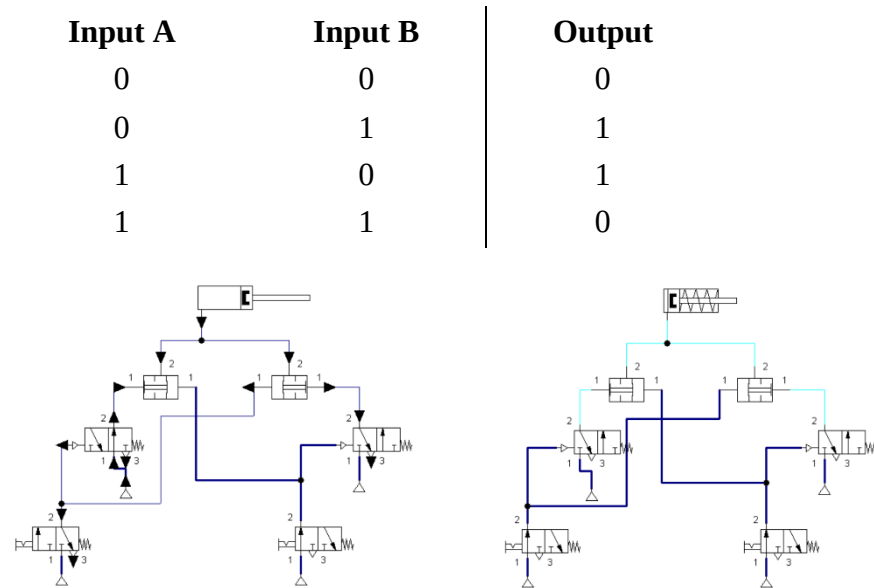


Figure 4.6: XOR Gate in (0,1) & (1,1) states

XNOR Gate

- Pneumatic Representation: Air flows only when both inputs are identical.
- Logical Behavior: Output is 1 when inputs are the same.

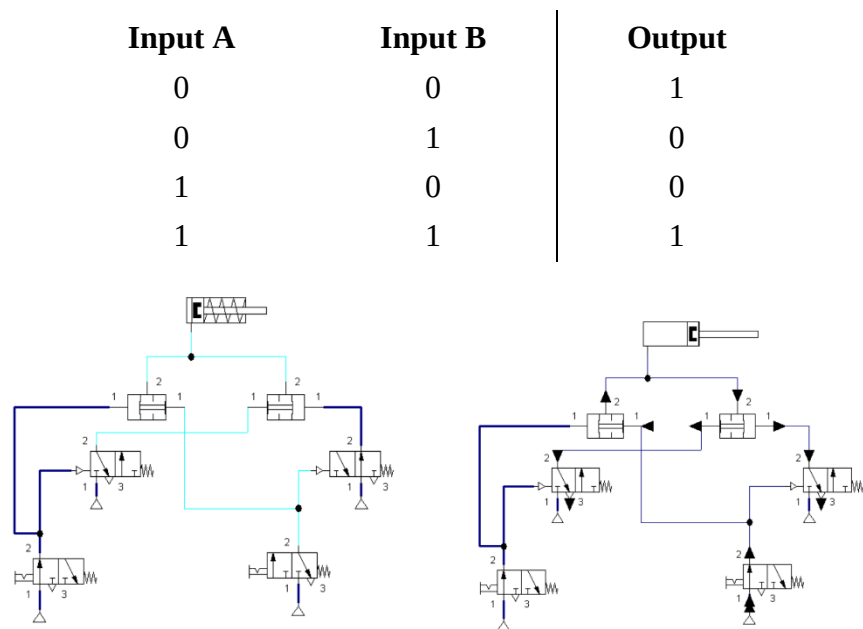


Figure 4.7: XNOR Gate in (1,0) & (1,1) states

Required Apparatus

- De Lorenzo Electro Pneumatic Trainer (DL 8115EPB)
- Single acting pneumatic cylinder
- Logic gate modules (AND, OR, NOT, NAND, NOR, XOR, XNOR)
- Push buttons
- 3/2-way solenoid valve
- Pneumatic tubing and connectors
- Air compressor
- 24V DC power supply
- Electrical wires

Circuit Buildup

OR Gate:

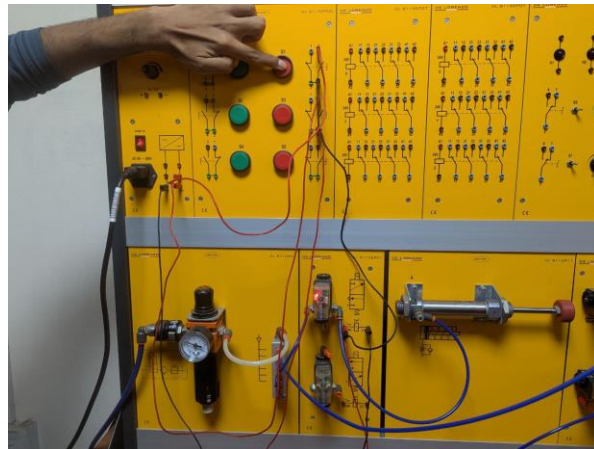


Figure 4.8: OR Logic Gate Buildup in DE LORENZO DL 8115EPB

AND Gate:

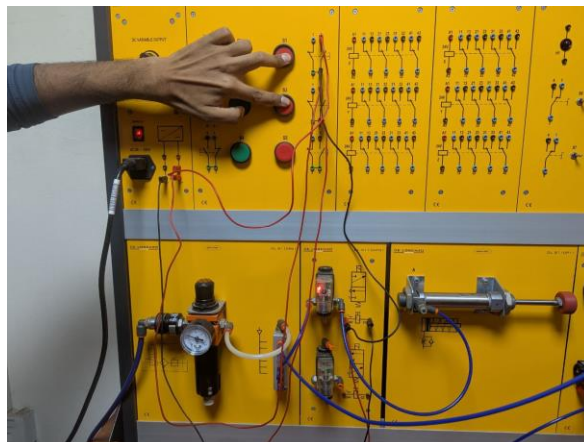


Figure 4.9: AND Logic Gate Buildup in DE LORENZO DL 8115EPB

NOT Gate:

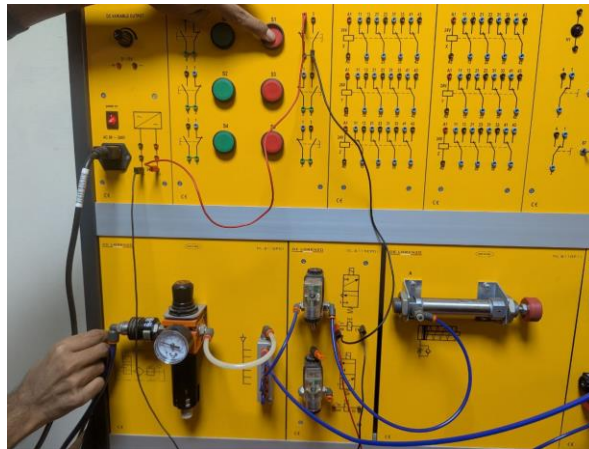


Figure 4.10: NOT Logic Gate Buildup in DE LORENZO DL 8115EPB

NAND Gate:



Figure 4.11: NAND Logic Gate Buildup in DE LORENZO DL 8115EPB

NOR Gate:



Figure 4.12: NOR Logic Gate Buildup in DE LORENZO DL 8115EPB

XOR Gate:

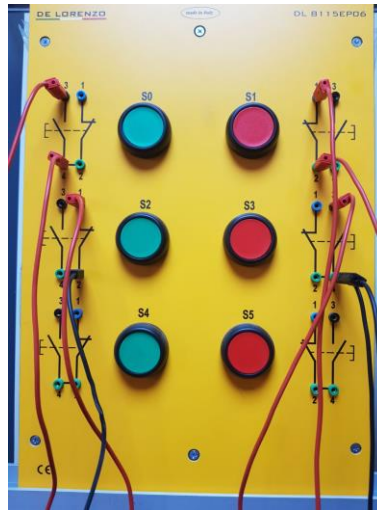


Figure 4.13: X-OR Logic Gate Buildup in DE LORENZO DL 8115EPB

XNOR Gate:



Figure 4.14: X-NOR Logic Gate Buildup in DE LORENZO DL 8115EPB

Applications

- Industrial Automation: For logic-driven machine actions
- Safety Systems: Ensuring safe operations using NOR/NAND logic
- Material Handling: Decision-based mechanisms using XOR/XNOR
- Educational Purposes: Understanding practical digital logic
- Simple Automation: Acts as a substitute for PLCs in budget-sensitive setups

Discussion

The experiment effectively demonstrated how logic gates can be used to control pneumatic actuators. Each logic condition was implemented and observed using the De Lorenzo trainer. Complex gates like XOR and XNOR showcased real-world control logic. Challenges such as circuit noise and timing delays were addressed, offering hands-on insight into digital control systems.

Conclusion

This experiment successfully showcased the integration of digital logic gates with a pneumatic actuator. By constructing various logic circuits, we were able to observe and understand the specific conditions required to control the motion of a single acting cylinder. These principles are foundational to automation, robotics, and industrial control systems.

References

- [1] "Single acting cylinder," THE INSTRUMENT GURU, 19-Jul-2022. [Online]. Available: <https://theinstrumentguru.com/single-acting-cylinder/>. [Accessed: 10-Apr-2025].
- [2] "Basic modular electro-pneumatics trainer," De Lorenzo Training & Didactical Equipment, 10-Apr-2024. Available: <https://delorenzoglobal.com/en/product/basic-electropneumatics-trainer/?Lang=En>. [Accessed: 10-Apr-2025].